Durant le tp, j'ai créé divers container, d'où les ID différents selon les captures, désolé par avance

```
 /lib/systemd/system/ubuntu-fan.service.
Paramétrage de docker.io (28.2.2-0ubuntu1~22.04.1) ...
Ajout du groupe « docker » (GID 136)...
Fait.
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /li
b/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /lib/sy
stemd/system/docker.socket.
Paramétrage de git (1:2.34.1-1ubuntu1.17) ...
Traitement des actions différées (« triggers ») pour man-db (2.10.2-1) ...
root@Ubuntu1:~# systemctl start docker
root@Ubuntu1:~#
```

```
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: >
     Active: active (running) since Wed 2026-03-18 08:10:31 CET; 1min 32s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 3236 (dockerd)
      Tasks: 8
     Memory: 22.2M
        CPU: 189ms
     CGroup: /system.slice/docker.service
             └─3236 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/contai>

mars 18 08:10:29 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:29.439014797+01:00>
mars 18 08:10:30 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:30.886426267+01:00>
mars 18 08:10:30 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:30.936767497+01:00>
mars 18 08:10:30 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:30.937150248+01:00>
mars 18 08:10:30 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:30.960032224+01:00>
[illegible] Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:30.960207587+01:00>
[illegible] Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:30.997957551+01:00>
mars 18 08:10:31 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:31.004480329+01:00>
mars 18 08:10:31 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:31.004627186+01:00>
mars 18 08:10:31 Ubuntu1 systemd[1]: Started Docker Application Container Engine.
~
~
```

Ubuntu Software

```
root@Ubuntu1:~# echo -e "[Service]\nEnvironment=\"HTTPS_PROXY=http://cache-etu.uni
v-artois.fr:3128/\"" | tee /etc/systemd/system/docker.service.d/http-proxy.conf
[Service]
Environment="HTTPS_PROXY=http://cache-etu.univ-artois.fr:3128/"
root@Ubuntu1:~# systemctl daemon-reload
root@Ubuntu1:~# systemctl status docker
○ docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: >
    Drop-In: /etc/systemd/system/docker.service.d
             └─http-proxy.conf
     Active: inactive (dead) since Wed 2026-03-18 08:16:39 CET; 5min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 3236 (code=exited, status=0/SUCCESS)
        CPU: 225ms

mars 18 08:10:30 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:30.997957551+01:00>
mars 18 08:10:31 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:31.004480329+01:00>
mars 18 08:10:31 Ubuntu1 dockerd[3236]: time="2026-03-18T08:10:31.004627186+01:00>
mars 18 08:10:31 Ubuntu1 systemd[1]: Started Docker Application Container Engine.
mars 18 08:16:39 Ubuntu1 systemd[1]: Stopping Docker Application Container Engine>
mars 18 08:16:39 Ubuntu1 dockerd[3236]: time="2026-03-18T08:16:39.193179851+01:00>
mars 18 08:16:39 Ubuntu1 dockerd[3236]: time="2026-03-18T08:16:39.196183769+01:00>
mars 18 08:16:39 Ubuntu1 systemd[1]: docker.service: Deactivated successfully.
mars 18 08:16:39 Ubuntu1 systemd[1]: Stopped Docker Application Container Engine.
mars 18 08:19:15 Ubuntu1 systemd[1]: /etc/systemd/system/docker.service.d/http-pr>
root@Ubuntu1:~# systemctl show --property=Environment docker
Environment=HTTPS_PROXY=http://cache-etu.univ-artois.fr:3128/
root@Ubuntu1:~#
```

```
root@Ubuntu1:~# systemctl start docker
root@Ubuntu1:~# docker run rtbethune/debian df -Th
Unable to find image 'rtbethune/debian:latest' locally
latest: Pulling from rtbethune/debian
df5590a8898b: Pull complete
f0556d053be8: Pull complete
Digest: sha256:09f58c01c1f8c74ecdcc013b2e2db34b957db9f43f1c597d29ff480660444e69
Status: Downloaded newer image for rtbethune/debian:latest
Filesystem     Type     Size  Used Avail Use% Mounted on
overlay        overlay  9.3G  8.9G     0 100% /
tmpfs          tmpfs     64M     0   64M   0% /dev
shm            tmpfs     64M     0   64M   0% /dev/shm
/dev/sda3      ext4     9.3G  8.9G     0 100% /etc/hosts
tmpfs          tmpfs    989M     0  989M   0% /proc/asound
tmpfs          tmpfs    989M     0  989M   0% /proc/acpi
tmpfs          tmpfs    989M     0  989M   0% /proc/scsi
tmpfs          tmpfs    989M     0  989M   0% /sys/firmware
tmpfs          tmpfs    989M     0  989M   0% /sys/devices/virtual/powercap
root@Ubuntu1:~#
```

docker ps

```
root@Ubuntu1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@Ubuntu1:~#
```

car le container debian a fini d'exécuter la commande démandée, comme plus aucun processus ne tourne à l'intérieur, Docker l'a arrêté

docker ps -a

```
root@Ubuntu1:~# docker ps -a
CONTAINER ID   IMAGE              COMMAND    CREATED              STATUS
              PORTS     NAMES
818703663dd0   rtbethune/debian   "bash"     About a minute ago   Exited (127) 44
seconds ago              debian
cb972fbceef4   rtbethune/debian   "df -Th"   24 minutes ago       Exited (0) 24 mi
nutes ago                sweet_payne
root@Ubuntu1:~#
```

l'option -a affiche l'historique, Docker conserve l'état du container même s'il n'est plus en cours d'exécution

On entre dans docker

```
root@Ubuntu1:~#
root@Ubuntu1:~# docker run -ti --name=debian rtbethune/debian bash
```

on passe alors de 'root@Ubuntu1' à 'root@ebb909e7a6f2' qui signifie qu'on est entré dans le container

version de debian

```
root@ebb909e7a6f2:/# cat /etc/os-release
PRETTY_NAME="Debian GNU/Linux 11 (bullseye)"
NAME="Debian GNU/Linux"
VERSION_ID="11"
VERSION="11 (bullseye)"
VERSION_CODENAME=bullseye
ID=debian
HOME_URL="https://www.debian.org/"
SUPPORT_URL="https://www.debian.org/support"
BUG_REPORT_URL="https://bugs.debian.org/"
root@ebb909e7a6f2:/# uname -a
```

version du noyau

```
root@ebb909e7a6f2:/# uname -a
Linux ebb909e7a6f2 5.15.0-43-generic #46-Ubuntu SMP Tue Jul 12 10:30:17 UTC 2022 x
86_64 GNU/Linux
root@ebb909e7a6f2:/#
```

@ip

```
root@ebb909e7a6f2:/# hostname -I
172.17.0.2
```

la commande ping ne semble pas être installée dans cette image debian très légère, donc cette commande ne semble pas faire partie du système de base du container
pour réussir le ping, il faudrait installer les outils réseaux dans le container via ‘apt install iputils-ping’, mais cela aurait alourdi l’image inutilement

```
root@ebb909e7a6f2:/# ping 8.8.8.8
bash: ping: command not found
```

création de fichier

```
root@ebb909e7a6f2:/# touch test
root@ebb909e7a6f2:/# ls
bin   dev  home  lib64  mnt  proc  run   srv  test  usr
boot  etc  lib   media  opt  root  sbin  sys  tmp   var
root@ebb909e7a6f2:/#
```

```
root@Ubuntu1:~# docker run -ti --name=debian rtbethune/debian bash
root@3728f2d66142:/#
```

```
vboxuser@Ubuntu1:~$ su -
Mot de passe :
root@Ubuntu1:~# docker ps
CONTAINER ID   IMAGE              COMMAND   CREATED          STATUS          PORTS
     NAMES
3728f2d66142   rtbethune/debian   "bash"    14 seconds ago   Up 13 seconds
     debian
root@Ubuntu1:~#
```

il y a bien 1 container actif, ID 3728f2d66142, son état est Up 13 seconds

création d’un fichier test, et usage commande ‘docker diff ID’

```
root@Ubuntu1:~# docker diff 3728f2d66142
A /test
root@Ubuntu1:~#
```

création du fichier docker-compose.yml

```
root@Ubuntu1:~# mkdir ~/test_apache && cd ~/test_apache
root@Ubuntu1:~/test_apache# nano docker-compose.yml
root@Ubuntu1:~/test_apache# cat docker-compose.yml
version: '3.3'

services:
  web:
    image: webdevops/php-apache:ubuntu-16.04
    container_name: web
    ports:
      - "80:80"
      - "443:443"
    volumes:
      - "$PWD/:/app/"
root@Ubuntu1:~/test_apache#
```

version : ‘3,3’ → version du format docker compose utilisée
services → début de la déclaration des containers à lancer
web → nom interne du service (servira de nom d’hôte sur le réseau docker)
image → image à télécharger sur le docker hub (ici apache avec php sous ubtunu 16,04)
container_name : web → force le nom du container à « web », plus facile pour le gérer ensuite
ports : - « 80:80 » → redirection de port : port de la vm : port du containers
volumes : - « $PWD/:/app/ » → le dossier actuel $PWD est « lié » au dossier /app du container, si je crées un fichier index.php sur la vm, il apparaît instantanément dans le serveur web

lancement de la commande docker-compose up -d

```
root@rt-mv:~# docker-compose up -d
Creating network "root_default" with the default driver
Pulling web (webdevops/php-apache:ubuntu-16.04)...
ubuntu-16.04: Pulling from webdevops/php-apache
fe703b657a32: Pull complete
f9df1fafd224: Pull complete
a645a4b887f9: Pull complete
57db7fe0b522: Pull complete
4c8b85719399: Pull complete
5bdbf4428cf9: Pull complete
61e7ce806845: Pull complete
aa6516eb9fc1: Pull complete
2dd7534a7ae2: Pull complete
0308fe8524ea: Pull complete
909cbb29bfc7: Pull complete
9ca0044427c7: Pull complete
942fd22230ad: Pull complete
270a36ce65b4: Pull complete
Digest: sha256:6f66066c604b0aa592a5cdfaddb6c0e0541aa54b642287155ffe2c37429fe1a2
Status: Downloaded newer image for webdevops/php-apache:ubuntu-16.04
Creating web ... done
root@rt-mv:~#
```

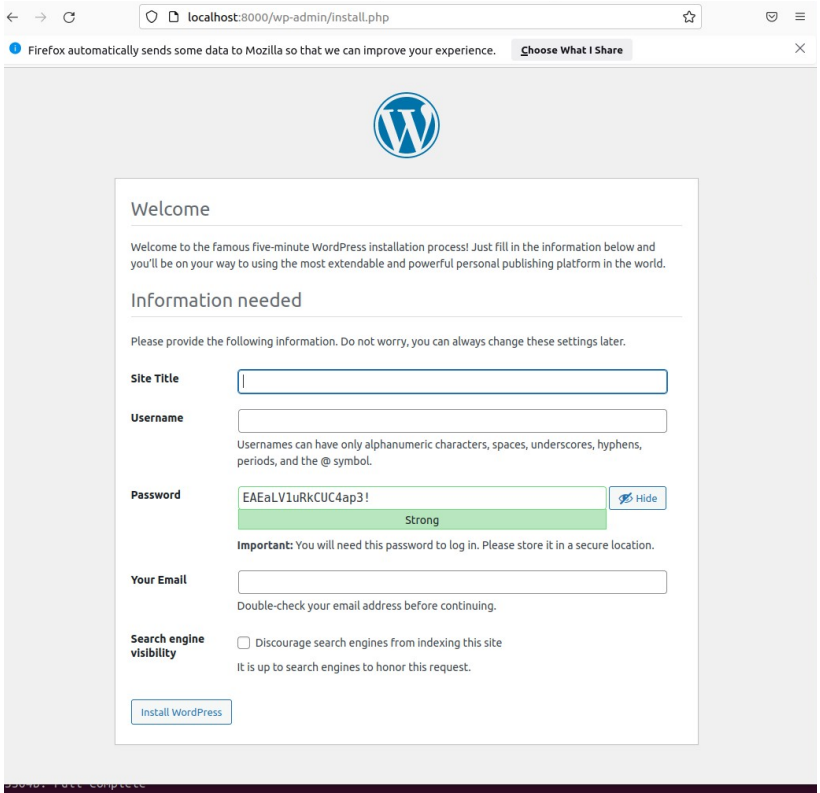
résultat de la commande commnande -help

```
root@rt-mv:~# docker-compose --help
Define and run multi-container applications with Docker.

Usage:
  docker-compose [-f <arg>...] [--profile <name>...] [options] [--] [COMMAND] [ARGS...]
  docker-compose -h|--help

Options:
  -f, --file FILE             Specify an alternate compose file
                              (default: docker-compose.yml)
  -p, --project-name NAME     Specify an alternate project name
                              (default: directory name)
  --profile NAME              Specify a profile to enable
  -c, --context NAME          Specify a context name
  --verbose                   Show more output
  --log-level LEVEL           Set log level (DEBUG, INFO, WARNING, ERROR, CRITICAL)
  --ansi (never|always|auto)  Control when to print ANSI control characters
  --no-ansi                   Do not print ANSI control characters (DEPRECATED)
  -v, --version               Print version and exit
  -H, --host HOST             Daemon socket to connect to

  --tls                       Use TLS; implied by --tlsverify
  --tlscacert CA_PATH         Trust certs signed only by this CA
  --tlscert CLIENT_CERT_PATH  Path to TLS certificate file
  --tlskey TLS_KEY_PATH       Path to TLS key file
  --tlsverify                 Use TLS and verify the remote
  --skip-hostname-check       Don't check the daemon's hostname against the
                              name specified in the client certificate
  --project-directory PATH    Specify an alternate working directory
                              (default: the path of the Compose file)
  --compatibility             If set, Compose will attempt to convert keys
                              in v3 files to their non-Swarm equivalent (DEPRECATED)
  --env-file PATH             Specify an alternate environment file

Commands:
  build              Build or rebuild services
  config             Validate and view the Compose file
  create             Create services
  down               Stop and remove resources
  events             Receive real time events from containers
  exec               Execute a command in a running container
  help               Get help on a command
  images             List images
  kill               Kill containers
  logs               View output from containers
  pause              Pause services
  port               Print the public port for a port binding
  ps                 List containers
  pull               Pull service images
  push               Push service images
  restart            Restart services
  rm                 Remove stopped containers
  run                Run a one-off command
  scale              Set number of containers for a service
  start              Start services
  stop               Stop services
  top                Display the running processes
  unpause            Unpause services
  up                 Create and start containers
  version            Show version information and quit
root@rt-mv:~#
```

ensuite on rouvre le fichier docker-compose.yml pour créer un blog wordpress avec une instance de bdd mysql

```
  GNU nano 6.2                                   docker-compose.yml *
version: '3.3'

networks:
   mon_reseau:
     driver: bridge
     ipam:
       config:
        - subnet: 10.16.0.0/24
volumes:
    dbdata:

services:
   db:
     image: mysql:5.7
     container_name: mysql
     volumes:
       - dbdata:/var/lib/mysql
     restart: always
     environment:
       MYSQL_ROOT_PASSWORD: somewordpress
       MYSQL_DATABASE: wordpress
       MYSQL_USER: wordpress
       MYSQL_PASSWORD: wordpress
     networks:
       - mon_reseau

   wordpress:
     depends_on:
       - db
     image: wordpress:latest
     container_name: wordpress
     ports:
       - "8000:80"
     restart: always
     environment:
       WORDPRESS_DB_HOST: db:3306
       WORDPRESS_DB_USER: wordpress
       WORDPRESS_DB_PASSWORD: wordpress
     networks:
       mon_reseau:
          ipv4_address: 10.16.0.100
```

on relance le server et démarre l’environnement

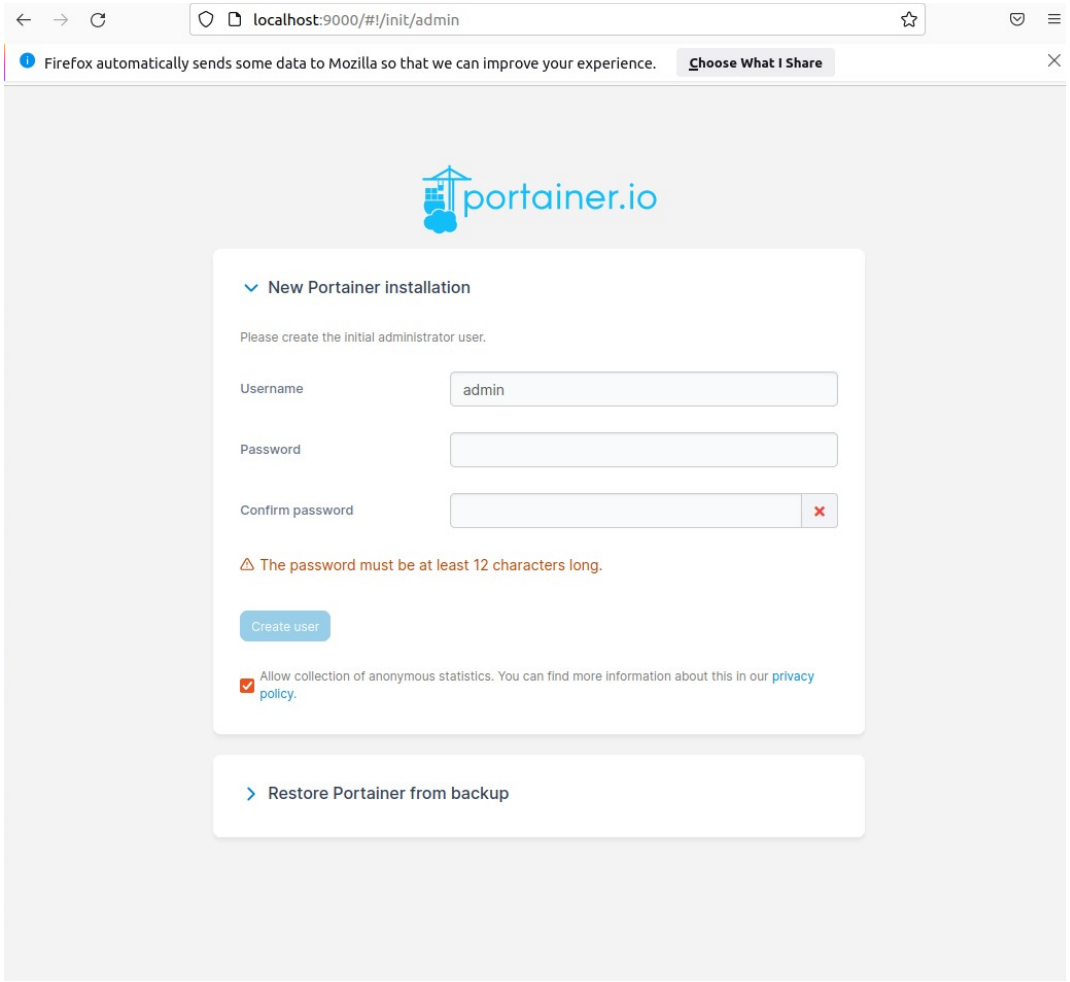


ça a marché

maintenant, on rouvre le fichier docker-compose.yml, on ajoute le container, on enregistre, et on relance la commande docker-compose up -d

```
  GNU nano 6.2                                    docker-compose.yml
version: '3.3'

networks:
   mon_reseau:
     driver: bridge
     ipam:
       config:
        - subnet: 10.16.0.0/24
volumes:
    dbdata:

services:
   db:
     image: mysql:5.7
     container_name: mysql
     volumes:
       - dbdata:/var/lib/mysql
     restart: always
     environment:
       MYSQL_ROOT_PASSWORD: somewordpress
       MYSQL_DATABASE: wordpress
       MYSQL_USER: wordpress
       MYSQL_PASSWORD: wordpress
     networks:
       - mon_reseau

   wordpress:
     depends_on:
       - db
     image: wordpress:latest
     container_name: wordpress
     ports:
       - "8000:80"
     restart: always
     environment:
       WORDPRESS_DB_HOST: db:3306
       WORDPRESS_DB_USER: wordpress
       WORDPRESS_DB_PASSWORD: wordpress
     networks:
       mon_reseau:
          ipv4_address: 10.16.0.100

   portainer:
     image: portainer/portainer
     container_name: portainer
     ports:
       - "9000:9000"
     volumes:
      - /var/run/docker.sock:/var/run/docker.sock
     networks:
       - mon_reseau
```

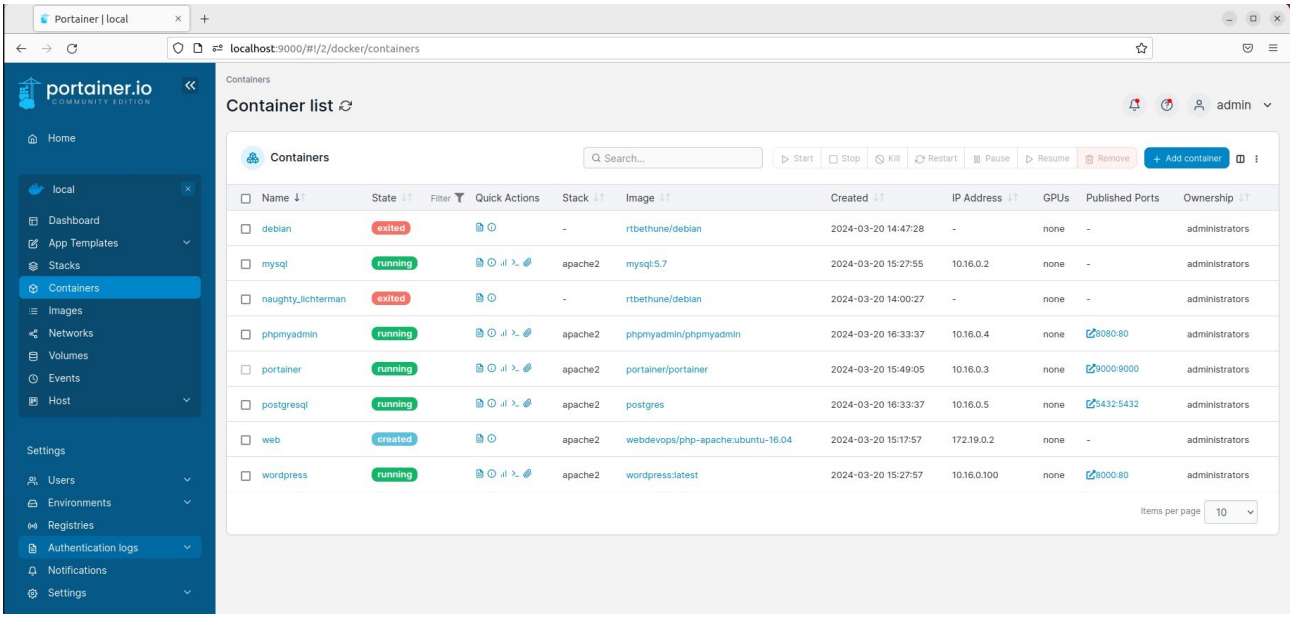
enfin on ouvre la page web sur http://localhost:9000



on modifie le fichier docker-composer.yml et on ajoute les services

```
  GNU nano 6.2                                                                  doc
      - mon_reseau

  wordpress:
    depends_on:
      - db
    image: wordpress:latest
    container_name: wordpress
    ports:
      - "8000:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wordpress
      WORDPRESS_DB_PASSWORD: wordpress
    networks:
      mon_reseau:
         ipv4_address: 10.16.0.100

  portainer:
    image: portainer/portainer
    container_name: portainer
    ports:
      - "9000:9000"
    volumes:
     - /var/run/docker.sock:/var/run/docker.sock
    networks:
      - mon_reseau

phpmyadmin:
    image: phpmyadmin
    container_name: phpmyadmin
    environment:
     - PMA_ARBITRARY=1
    restart: always
    ports:
     - 8080:80
    volumes:
     - /sessions

# Use postgres/example user/password credentials

services:

  db:
    image: postgres
    restart: always
    # set shared memory limit when using docker compose
    shm_size: 128mb
    # or set shared memory limit when deploy via swarm stack
    #volumes:
    #  - type: tmpfs
    #    target: /dev/shm
    #    tmpfs:
    #      size: 134217728 # 128*2^20 bytes = 128Mb
    environment:
      POSTGRES_PASSWORD: example

  adminer:
    image: adminer
    restart: always
    ports:
      - 8080:8080

^G Aide        ^O Écrire      ^W Chercher    ^K Couper      ^T Exécuter    ^C Emplacement  M-U Annuler    M-A Marquer
^X Quitter     ^R Lire fich.  ^\ Remplacer   ^U Coller      ^J Justifier   ^/ Aller ligne   M-E Refaire    M-6 Copier
```

on relance le server




```
administrateur@rt-mv:~$ sudo -s
[sudo] Mot de passe de administrateur :
root@rt-mv:/home/administrateur# docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED             STATUS              PORTS
                                                                                   NAMES
472ee693b283   postgres               "docker-entrypoint.s…"   2 minutes ago       Up 2 minutes        0.0.0
.0:5432->5432/tcp, :::5432->5432/tcp                                               postgresql
e6770706c663   phpmyadmin/phpmyadmin  "/docker-entrypoint.…"   2 minutes ago       Up 2 minutes        0.0.0
.0:8080->80/tcp, :::8080->80/tcp                                                   phpmyadmin
d7a723e83b9e   portainer/portainer    "/portainer"             47 minutes ago      Up 47 minutes       8000/
tcp, 9443/tcp, 0.0.0.0:9000->9000/tcp, :::9000->9000/tcp                           portainer
03b0efbf9349   wordpress:latest       "docker-entrypoint.s…"   About an hour ago   Up About an hour    0.0.0
.0:8000->80/tcp, :::8000->80/tcp                                                   wordpress
a571ce197815   mysql:5.7              "docker-entrypoint.s…"   About an hour ago   Up About an hour    3306/
tcp, 33060/tcp                                                                     mysql
root@rt-mv:/home/administrateur#
```